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LASCAS: TOWARD COGNITIVE ACCESSIBILITY IN LEGAL TEXTS
Legal Accessibility and Simplification for Cognitive Aid System

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LASCAS: TOWARD COGNITIVE ACCESSIBILITY IN LEGAL TEXTS

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Abstract: Legal documents are fundamental to civic participation, yet their inherent complexity creates significant barriers to comprehension, particularly for individuals with cognitive and learning disabilities. This study addresses legal text inaccessibility by proposing LASCAS (Legal Accessibility and Simplification for Cognitive Aid System), a local-first prototype designed to simplify Brazilian legal Portuguese. The work is grounded in cognitive accessibility principles, the Plain Language movement, and Automatic Text Simplification (ATS), combining modern NLP techniques with semantic-integrity safeguards and accessibility-oriented presentation. Empirically, a formative pre-test with 3 legal-domain participants supported the adoption of the medium simplification level, and a broader questionnaire study with 30 participants produced consistently positive results: Perceived Ease of Use and Learnability both averaged 4.68/5, Perceived Usefulness 4.63/5, and the SUS-style composite 90.4/100, with 86.7% of participants at 80 or above. A backend benchmark on consumer-grade hardware (8 GB RTX 4070 Laptop GPU) simplified a 4,392-word petition at the medium level in 153.5 s on average (about 35 s per 1,000 words) with near-saturated GPU utilization, evidencing the feasibility and hardware sensitivity of the local-first design. These results suggest that the prototype is already effective as a support tool for the initial reading, comprehension, and triage of legal documents, and that LASCAS is a promising approach to advancing access to justice and digital inclusion.

Keywords: Text Simplification. Cognitive Accessibility. Natural Language Processing. Legal Tech. Transformer Models.

1 INTRODUCTION

Recent advances in large language models and Natural Language Processing (NLP) are creating increasingly sophisticated systems capable of processing complex legal language and supporting a growing range of legal tasks (HE et al., 2026). This technological advancement holds particular promise for the legal domain, where documents ranging from judicial rulings to terms of service form the bedrock of civic life. However, a profound disconnect exists between the importance of these texts and their accessibility. The specialized language of law, or "Legalese," is characterized by an intricate vocabulary and convoluted sentence structures, creating a formidable barrier to comprehension for non-experts (LAI et al., 2024).

The scale of this challenge is amplified by literacy statistics. In Brazil, data indicates that a significant portion of the population lacks the functional literacy skills required to parse complex, high-stakes documents (ALVES et al., 2023). This gap is not merely academic; it has profound consequences. The inability to understand a contract or a court summary impedes an individual's ability to make informed decisions and exercise their rights, creating a depen-

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dency on legal professionals whose services can be prohibitively expensive. This exacerbates social inequities and limits access to justice, a problem that automated simplification systems are uniquely positioned to address (AGRAWAL; CARPUAT, 2024).

This problem of inaccessible legal language is a recognized societal challenge, prompting action from governmental bodies worldwide. Initiatives such as the Plain Writing Act of 2010 in the United States and resolutions from Brazil's National Council of Justice (CNJ) explicitly call for clearer communication (ALVES et al., 2023). The development of automated systems for legal text simplification is, therefore, a direct response to a pressing need to bridge the communication gap between the judiciary and the populace. Such technology aligns with a global movement towards making legal information a public good, accessible to all regardless of educational background.

The response to this challenge is increasingly found at the intersection of law and artificial intelligence, often termed "Legal Tech" (PORTO; ARAÚJO; GABRIEL, 2024). Specifically, the field of Natural Language Processing (NLP) offers the tools to analyze and, more importantly, transform complex legal language at scale (ARIAI; MACKENZIE; DEMARTINI, 2025). This technological intervention is not merely for convenience; it is a necessary step to operationalize the principles of cognitive accessibility and Plain Language for the vast and growing body of digital legal information. This work is situated within this exact context, proposing a computational approach to mitigate a deeply rooted communication barrier.

1.1 Problem Statement

The core problem addressed by this thesis is that legal texts and judicial communications in Brazilian Portuguese often require simplification so that legal and judicial information can be communicated in simple, direct, and comprehensible language to all citizens (Conselho Nacional de Justiça, 2023). This complexity is not a mere stylistic inconvenience; it functions as a systemic barrier that can reinforce dependence on legal intermediaries and make access to legal information less autonomous. Recent scholarship on legislative drafting similarly argues that laws intended for broad public audiences must be written with understandability and accessibility in view, rather than primarily for specialist readers (ZARISKI, 2026).

From an accessibility perspective, text simplification is relevant because it can help people with cognitive impairments, language learners, and children understand complex content, but readability improvements alone are not sufficient if the simplified text fails to preserve grammatical adequacy and meaning (ALISSA; WALD, 2023).

Consequently, this inaccessibility undermines equal access to legal information and informed decision-making in digital contexts, where users are often expected to accept complex legal terms that are difficult for ordinary readers to understand (BENOLIEL; BECHER, 2019). This point is critical in the current technological landscape: as essential civic and commercial interactions move online, this linguistic barrier reflects a broader form of digital inequality,

where access to digital platforms is not enough if users lack the skills, conditions, or support needed to make effective use of online information and services (LYTHREATIS; SINGH; EL-KASSAR, 2022).

1.2 Research Question and Objectives

This study is guided by a central research question and a set of objectives designed to systematically explore the problem space and potential solutions.

Research Question: How can modern Natural Language Processing techniques, particularly those based on the Transformer architecture, be effectively applied to simplify Brazilian Portuguese legal texts to improve their cognitive accessibility in accordance with Plain Language principles?

Objectives:

- To establish the conceptual and technical foundations of LASCAS by relating cognitive accessibility, Plain Language, and Automatic Text Simplification in the context of Brazilian Portuguese legal documents.
- To develop the LASCAS prototype as a local-first architecture capable of simplifying real Brazilian legal documents (HTML and PDF) while preserving essential legal information.
- To evaluate the prototype in terms of perceived understanding, usability, and its main strengths and limitations for improving access to legal information.

1.3 Division of Work

This thesis progresses from foundations to implementation and evaluation. Section I presents the conceptual background and related work. Section II describes the architecture and development of the LASCAS prototype. Section III reports the evaluation, discusses the results, and highlights limitations and future work.

2 BACKGROUND

To situate the LASCAS project, this section outlines the foundational concepts at the intersection of accessibility, linguistics, and computation. We first explore the "why"—the principles of cognitive accessibility and Plain Language that define the goal. We then detail the "how"—the core technologies of Automatic Text Simplification (ATS) and the Transformer models that enable it. The specific features that define the complexity of legal text, which this project directly targets, are well-documented and can be categorized as follows:

- *Lexical Complexity*: The pervasive use of domain-specific jargon, archaic terminology, and foreign phrases unfamiliar to non-experts.

- *Syntactic Complexity*: The prevalence of long, convoluted sentences containing multiple nested clauses and the passive voice, which can obscure meaning and increase processing difficulty.
- *High Information Density*: The condensation of numerous conditions and exceptions into single, dense paragraphs, demanding high levels of sustained attention and working memory.

2.1 Cognitive Accessibility and Plain Language

Cognitive accessibility is an aspect of inclusive design focused on making information understandable for people with a wide range of cognitive and learning disabilities affecting memory, attention, and language comprehension (World Wide Web Consortium, 2023). The W3C's Web Content Accessibility Guidelines (WCAG) provide a framework for this, with Guideline 3.1, "Readable," being particularly pertinent. Its intent is to ensure users can correctly interpret text. Key success criteria under this guideline that directly motivate this thesis include (World Wide Web Consortium, 2023):

- *Unusual Words (3.1.3)*: Recommends providing a mechanism for identifying definitions of words or phrases used in an unusual or restricted way, including jargon and idioms.
- *Abbreviations (3.1.4)*: Calls for a mechanism to be available for identifying the expanded form or meaning of abbreviations.
- *Reading Level (3.1.5)*: Considered a cornerstone of this work, this criterion recommends that when content requires an advanced reading ability, a supplementary version that does not require such ability should be provided.

Collectively, these criteria establish a direct mandate for text simplification as a necessary accommodation. While cognitive accessibility defines the goal, the Plain Language framework provides the methodology. Plain Language is communication so clear that the intended audience can easily find what they need, understand it, and use it effectively. Its principles offer concrete guidelines for rewriting complex text, which can be operationalized by an automated system (International Organization for Standardization, 2023).

Key principles include structural simplicity (short sentences, clear headings), lexical simplicity (common words instead of jargon), and directness (using the active voice). Manually applying this methodology to the vast corpus of legal documents is impractical (ALVES et al., 2023). This is where NLP serves as the enabling tool, providing the means to automate the Plain Language methodology at scale. This thesis is therefore positioned at the intersection of these fields, aiming to operationalize a design philosophy (Plain Language) with a specific technology (NLP) to achieve a critical goal (Cognitive Accessibility).

2.2 Automatic Text Simplification with Transformers

Natural Language Processing (NLP) is a subfield of AI concerned with enabling computers to understand and manipulate human language. Within NLP, Automatic Text Simplification (ATS) aims to transform a text into an easier-to-read version while preserving its original meaning (ALVES et al., 2023). It is distinct from summarization, which inherently reduces information content. ATS operates through two primary sub-tasks: lexical and syntactic simplification.

The majority of state-of-the-art NLP models are based on the Transformer architecture (ALISSA; WALD, 2023). The Transformer’s key innovation was its reliance on attention mechanisms, which proved highly effective at capturing long-range dependencies between words. It follows an encoder-decoder structure well-suited for text-to-text tasks like simplification. The encoder processes the input complex sentence to build a context-aware numerical representation, and the decoder uses this representation to generate the simplified output one token at a time.

The core of this architecture is the multi-head self-attention mechanism, which allows the model to weigh the importance of different words in the input when processing the sequence. This enables it to jointly attend to information from different representational subspaces, enhancing its ability to capture the complex relationships between words in a sentence (ALISSA; WALD, 2023). This architecture provides the powerful foundation upon which modern simplification models are built.

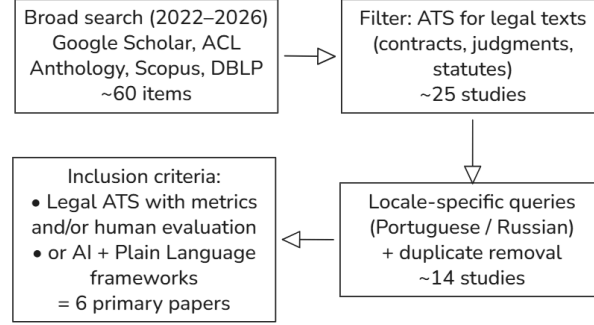
3 RELATED WORK

Related work was identified through a reproducible, funnel-style search across Google Scholar, ACL Anthology, Scopus, and DBLP. A broad search combining the terms “legal text simplification,” “plain language,” and “law/legal,” with a 2022–current year filter, yielded ~ 60 items screened by title and abstract. Scope was then narrowed to *Automatic Text Simplification* (ATS) in the legal domain through combinations of “text simplification,” “legal,” and document types such as contracts, judgments, and statutes, alongside model terms such as Transformer, T5, and BART, reducing the pool to ~ 25 .

Locale-sensitive searches also targeted Portuguese and Russian efforts. After removing duplicates and near-duplicates, ~ 14 studies remained. Inclusion then required either empirical ATS on legal data with metrics and/or human judgments, or peer-reviewed conceptual frameworks grounding plain-language and AI-assisted rephrasing for legal accessibility. After full-text screening, 6 primary papers remained. The discussion below focuses on three empirical ATS studies in the legal domain—(GARIMELLA et al., 2022; ALVES et al., 2023; ATHUGODAGE; MITROFANOVA; GUDKOV, 2024)—and two works framing plain language and AI-assisted rephrasing as accessibility enablers in law—(TOIVONEN; KETOLA; HAAPIO, 2025; ZIMINA-POIROT; GLEDHILL; BOUYÉ, 2025). This set aligns with the identified

research gap (semantic fidelity vs. accessibility under data scarcity). Figure 1 summarizes this search and screening process.

Figure 1: Funnel-style search and screening process for related work.



Source: prepared by the author (2026)

(GARIMELLA et al., 2022) explored text simplification (TS) for legal contracts, highlighting the lack of parallel data as a key challenge. By evaluating unsupervised methods and a supervised BART model, they found significant *disagreement among legal experts on simplicity* and *weak correlations between automatic metrics and human judgments* (GARIMELLA et al., 2022). Their work underscored the trade-offs between fluency, meaning preservation, and simplification level.

(ALVES et al., 2023) focus on Brazilian Portuguese *judicial summaries*, tackling data scarcity by generating a synthetic parallel dataset with the unsupervised MUSS model (ALVES et al., 2023). This data was then used to train supervised Transformer and NMT models. Their evaluation showed that most methods, especially the Transformer model and MUSS combined with translation, *improved Flesch Reading Ease (FRE) scores* over the original summaries.

(ATHUGODAGE; MITROFANOVA; GUDKOV, 2024) address data scarcity for Russian legal text by creating a parallel corpus that pairs official legal documents with plain-language explanations from the "Rossiyskaya Gazeta" newspaper (ATHUGODAGE; MITROFANOVA; GUDKOV, 2024). Using this dataset, they *fine-tuned T5 and GPT models* which, despite poor ROUGE scores, achieved good results in SARI and readability metrics and often exceeded human-written commentaries in human evaluation.

(TOIVONEN; KETOLA; HAAPIO, 2025) discuss plain-language communication and the role of AI within the emerging paradigm of “personalized law,” arguing that user-centered legal design and clear wording are prerequisites for accessibility rather than optional stylistic choices (TOIVONEN; KETOLA; HAAPIO, 2025). They caution that AI should augment—not replace—human judgment, recommending standards and oversight to avoid opacity and loss of legal nuance. Although not an empirical simplification benchmark, the paper positions plain language and responsible AI as foundations for scalable legal accessibility.

(ZIMINA-POIROT; GLEDHILL; BOUYÉ, 2025) examine generative AI models as con-

tent rephrasing assistants for producing Plain Language and Easy-to-Read versions of legal texts in multilingual settings (ZIMINA-POIROT; GLEDHILL; BOUYÉ, 2025). Their case-based analysis shows that, although GAI can substantially simplify phrasing and structure, it still lacks functional legal awareness; consequently, effective workflows require human oversight, clear domain guidelines, and review tools to prevent semantic and pragmatic losses.

Overall, the literature evolves from exposing the limits of existing legal TS methods to proposing data-creation strategies and, more recently, broader plain-language and human-in-the-loop perspectives. Across these studies, the shared challenge remains improving readability without compromising legal meaning, especially in data-scarce settings. This gap directly motivates the LASCAS proposal.

4 LASCAS MODEL

To address the problem of legal text inaccessibility, this thesis proposes the conceptualization and development of the LASCAS (Legal Accessibility and Simplification for Cognitive Aid System) prototype. LASCAS is envisioned as an Automatic Text Simplification (ATS) system that leverages modern Natural Language Processing (NLP) techniques to transform complex legal Brazilian Portuguese into a more understandable form aligned with Plain Language principles. The primary goal of LASCAS is to enhance the cognitive accessibility of legal documents by systematically reducing their lexical and syntactic complexity, thereby lowering the cognitive load required for comprehension.

4.1 Architecture

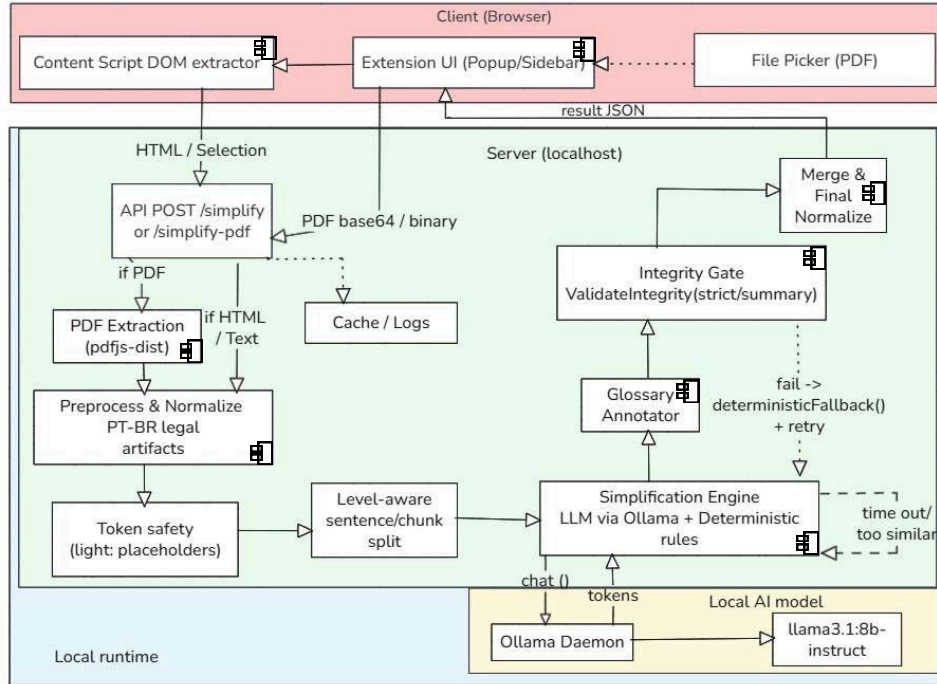
LASCAS is a local-first pipeline that ingests HTML or PDF, normalizes legal artifacts, applies **level-aware token safety**, and then simplifies the text using a hybrid core that combines a local LLM (via Ollama) with deterministic fallbacks. A thin browser extension invokes the local API and renders results side-by-side. The main components are:

- *Local API (Node/Express)*: the primary JSON endpoint `/simplify` orchestrates the end-to-end flow for HTML and PDF (via URL or Base64). In addition, the prototype exposes `/simplify-pdf` for binary PDF uploads (`application/pdf`), which reduces payload overhead for large files and improves robustness in real usage. The same server layer also centralizes lightweight cache/log traces used for request inspection, timing analysis, and local debugging during development and evaluation.
- *PDF Extraction*: robust text extraction using `pdfjs-dist` legacy build; includes a geometry-based reconstruction that re-assembles lines by x - y coordinates and adaptive gap heuristics, plus page-gluing rules that avoid spurious blank paragraphs.

- *Preprocessing*: normalization of legal artifacts (numbers, punctuation, diacritics, currency, dates), sentence splitting with abbreviation shielding, and paragraph-aware chunking for the condensation levels. Token safety is level-aware: in `light`, placeholders are used to protect critical items before any LLM call. In `medium/strong`, the system prioritizes the **preservation and validation** of essential items (e.g., dates, values, percentages, legal citations, registry/process identifiers, and negations), while PII/qualification fields may be omitted by design.
- *Simplification Core*: hybrid engine that (i) prompts the local model llama3.1:8b-instruct via Ollama (configured in the prototype through `LASCAS_MODEL`); (ii) rejects unusable outputs (timeouts, meta-preambles, or overly similar rewrites); (iii) deterministic restructuring fallback that enforces visible structure (short lines, headings, bullets); and (iv) glossary first-occurrence annotations for unusual words and abbreviations (WCAG 3.1 intent).
- *Merge and final normalization*: after sentence- or chunk-level processing, accepted outputs are rejoined and post-normalized before the response is returned to the browser. This stage consolidates intermediate results, restores protected content when applicable, and stabilizes headings, spacing, and final presentation.
- *Integrity Layer (level-aware)*: semantic safety check that validates preservation of essential legal information. In `light`, integrity is strict (high fidelity, near-complete preservation). In `medium/strong`, integrity is configured for summarization: it still preserves dates, values, percentages, legal citations and registry/process references, but allows non-essential personal qualifications (e.g., CPF/RG, address, repetitions) to be dropped to achieve meaningful condensation.
- *Extension Shell (MV3)*: the popup selects the simplification level and coordinates browser-side inputs through the content script, including DOM/selection extraction and picked PDF files; CSS sidebar is injected as a web-accessible resource for side-by-side viewing. PDFs can be forwarded as Base64 through `/simplify` or as binary uploads through `/simplify-pdf`, depending on browser constraints and file size.

Figure 2 summarizes the high-level architecture of LASCAS, including the user interface, local service, and document sources. The diagram should be read from the browser layer to the local runtime layer: user input enters through the extension, is routed to the local API, and is then processed by extraction, normalization, simplification, integrity validation, and final presentation stages. This organization reinforces the central design decision of the prototype: the language model is not isolated from the rest of the system, but embedded in a controlled workflow that prepares the input before generation and validates the output before returning it to the user.

Figure 2: Component-level architecture



Source: prepared by the author (2026)

Component mapping to files. Table 1 maps each logical component of the architecture to its corresponding source files.

Table 1: Component mapping to source files in LASCAS.

Component	File(s)
Local API / warm-up / route	lascas-local/server.js
PDF extractor & fallbacks	lascas-local/src/pdf.js
Preprocess & protection	lascas-local/src/preprocess.js
Glossary annotations	lascas-local/src/glossary.js
Hybrid simplification core / final normalization	lascas-local/src/simplifier.js
Integrity validator	lascas-local/src/validators.js
MV3 manifest	lascas-extension/public/manifest.json
Content CSS	lascas-extension/public/sidebar.css
Popup UI / PDF picker (React)	lascas-extension/src/App.tsx
Popup bootstrap (React)	lascas-extension/src/main.tsx
Content script / DOM extraction	lascas-extension/src/content.ts

Source: prepared by the author (2026)

4.2 Algorithm

The LASCAS pipeline is implemented as a level-aware procedure. The `light` level focuses on sentence-level rewriting (high fidelity, near-original length), while `medium` and `strong` apply condensation with a guaranteed-shrink heuristic fallback. Algorithm 1 presents the step-wise procedure used in the prototype.

More specifically, the algorithm makes explicit that simplification is not treated as a single generation step. The pipeline first normalizes and pre-structures the input, then branches according to the selected level. In `light`, the system protects sensitive content, rewrites sentence by sentence, restores protected items, and finally applies a strict integrity check.

In `medium` and `strong`, the system works on paragraph-aware chunks, selects the legal elements that must remain traceable, and only accepts a condensed candidate when it satisfies both structural and semantic constraints. If the model output is empty, too similar to the source, excessively verbose, or fails the preservation checks, the procedure falls back to deterministic condensation or restructuring. This sequence clarifies how LASCAS combines generation, validation, and recovery within the same processing flow, so that accessibility gains are pursued without giving the model uncontrolled authority over legally relevant content.

The main hyperparameters for each simplification level are summarized in Table 2. These parameters are not merely implementation details: they encode the expected behavior of each level by controlling whether the system should rewrite or condense, how much variation the model may introduce, how aggressively the output should shrink, and how long each local call may run before fallback behavior becomes preferable.

Table 2: Level parameters used by the prototype backend.

Level	Mode	Temperature	Output target	Per-call budget (ms)
light	rewrite (sentence)	0.12	$\approx 0\text{--}20\%$	24,000
medium	condense (chunk)	0.16	target ratio ≈ 0.55	52,000
strong	condense (chunk)	0.18	target ratio ≈ 0.35	65,000

Source: prepared by the author (2026)

For condensation levels, paragraph-aware chunking uses 520–1050 words per chunk, and a one-time cold-start boost is applied to early calls to reduce timeout risk.

Algorithm 1: LASCAS–Level–Aware–Simplify

Input : T_{in} (raw text), $level \in \{\text{light}, \text{medium}, \text{strong}\}$
Output: T_{out} simplified

```

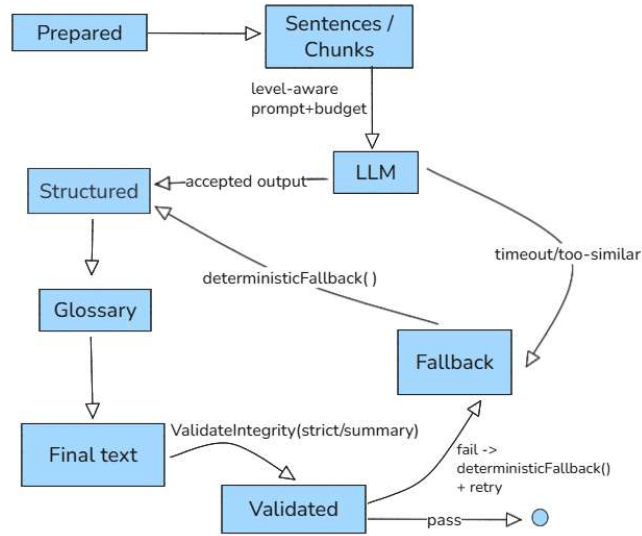
1  $T_0 \leftarrow \text{NormalizeArtifactsForLegal}(T_{\text{in}})$ 
2  $T_1 \leftarrow \text{DeterministicPreStructure}(T_0)$ 
3 if  $level = \text{light}$  then
  // High-fidelity sentence rewrite
4  $(T_{\text{prot}}, PH) \leftarrow \text{ProtectPlaceholdersLight}(T_1)$ 
5  $P \leftarrow \text{SplitParagraphs}(T_{\text{prot}})$ 
6  $Out \leftarrow []$ 
7 foreach  $p \in P$  do
8   if  $\text{IsHeadingLike}(p)$  then // Keep headings as-is
9      $\text{Append}(Out, p)$ ; continue
10   $S \leftarrow \text{SplitSentencesProtected}(p)$ 
11   $S' \leftarrow []$ 
12  foreach  $s \in S$  do
13     $cand \leftarrow \text{TryCallOllamaRewrite}(s, \text{Budget}(level))$ 
14    if  $\text{AcceptRewrite}(s, cand)$  then
15       $\text{Append}(S', cand)$ 
16    else
17       $\text{Append}(S', s)$ 
18   $\text{Append}(Out, \text{Join}(S', ""))$ 
19  $T_{\text{prot}} \leftarrow \text{Join}(Out, "\n\n")$ 
20  $T_{\text{out}} \leftarrow \text{UnprotectPlaceholders}(T_{\text{prot}}, PH)$ 
21  $T_{\text{out}} \leftarrow \text{PostStructureAndNormalize}(T_{\text{out}})$ 
22  $T_{\text{out}} \leftarrow \text{AnnotateFirstOccurrences}(T_{\text{out}})$ 
23 if not  $\text{ValidateIntegrity}(T_0, T_{\text{out}}, \text{strict})$  then
24    $T_{\text{out}} \leftarrow \text{DeterministicFallback}(T_0)$ 
25 else
  // Condensation with guaranteed shrink
26  $C \leftarrow \text{SplitIntoChunksByWords}(T_1, 520, 1050)$ 
27  $Out \leftarrow []$ 
28 foreach  $c \in C$  do
29    $req \leftarrow \text{PickRequiredTokens}(c, level)$ 
30    $cand \leftarrow \text{TryCallOllamaCondense}(c, req, \text{Budget}(level))$ 
31   if  $\text{AcceptCondense}(c, cand, req)$  then
32      $\text{Append}(Out, cand)$ 
33   else
34      $h \leftarrow \text{HeuristicCondense}(c, level)$ 
35      $\text{Append}(Out, h)$ 
36  $T_{\text{out}} \leftarrow \text{PostStructureAndNormalize}(\text{Join}(Out, "\n\n"))$ 
37  $T_{\text{out}} \leftarrow \text{AnnotateFirstOccurrences}(T_{\text{out}})$ 
38 if not  $\text{ValidateIntegrity}(T_0, T_{\text{out}}, \text{summary})$  then
39    $T_{\text{out}} \leftarrow \text{DeterministicFallback}(T_0)$ 
40 return  $T_{\text{out}}$ 

```

Source: prepared by the author (2026)

Figure 3 illustrates the control logic and decision states for a single simplification pass. It complements Algorithm 1 by showing, in a more visual form, where the pipeline branches, how candidate outputs are accepted or rejected, and at which points fallback behavior is triggered. In this sense, the figure helps clarify that simplification in LASCAS is governed by successive decision checks rather than by a single opaque model response.

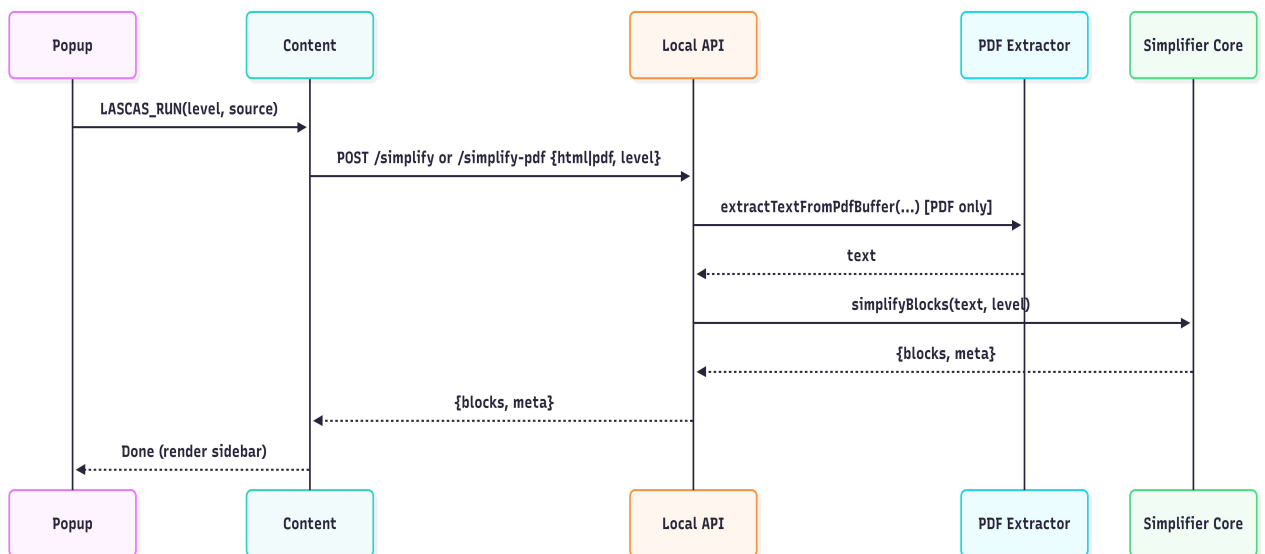
Figure 3: Decision states for a single pass.



Source: prepared by the author (2026)

Figure 4 shows the end-to-end request path between the browser extension and the local LASCAS service. While the architectural diagram identifies the main components, this sequence view clarifies how they interact during actual use: the extension captures the user input, sends the request to the local backend, waits for extraction and simplification to finish, and then receives the final result for presentation in the browser. This makes the operational coordination of the prototype more explicit.

Figure 4: End-to-end request path



Source: prepared by the author (2026)

4.3 Aspects of Implementation

This section details key technical choices and components within the LASCAS prototype, focusing on robustness, legal-specific processing, safety mechanisms, and integration points.

- *PDF robustness*: The extractor uses `pdfjs-dist` legacy build with explicit worker, CMAP paths and applies a geometry-based reconstruction that re-assembles lines by *y*-banding and adaptive gap thresholds (percentiles of glyph gaps per line). It also fixes **dash-line breaks** in numbers (`1-\n2` $\rightarrow$ `1-2`), collapses excessive newlines, and glues page breaks using sentence-aware heuristics to avoid accidental paragraph splits.
- *Legal-aware normalization and token safety (level-aware)*: The normalizer repairs Brazilian formats R\$ 133.569,32, digit spacing artifacts (`133.569, 32` $\rightarrow$ `133.569,32`), and OCR-style issues typical of notary deeds. Token safety is level-aware: (i) in `light`, placeholders are used to protect critical content before LLM calls (numbers, dates, values, legal citations, negations, and fragile legal terms) and are restored after rewriting; (ii) in `medium/strong`, the system focuses on preserving and validating **essential** items (dates, values, percentages, legal citations, and registry/process identifiers), while allowing non-essential qualification/PII fields to be omitted to achieve meaningful condensation.
- *Sentence splitting and chunking*: Abbreviations (e.g., **Art.**, **Dr.**, **Av.**) are shielded to avoid spurious splits. In `light`, rewriting is performed sentence-by-sentence within paragraphs. In `medium/strong`, the input is split into paragraph-aware chunks (520–1050 words) to respect token budgets and latency caps.
- *LLM adapter and latency control*: The adapter targets a local Ollama endpoint and prompts `llama3.1:8b-instruct` (set through `LASCAS_MODEL` in the prototype setup). It computes a dynamic `num_predict` from input size and the per-level target, enforces per-call budgets, and uses soft timeouts (via abort signals). Cold starts receive a one-time boost for the first calls. Outputs that are empty, contain meta-preambles, or fail acceptance checks (e.g., insufficient change for rewrite; insufficient shrink for condensation) are rejected.
- *Deterministic structuring*: The fallback enforces visible structure: short lines, stable headings, bulletization of semicolon-delimited lists, and targeted legal phrasing fixes. This guarantees readability even without the LLM and reduces the risk of returning nearly unchanged legalese.
- *Glossary and cognitive accessibility*: On the first occurrence, selected legal terms are annotated inline with simple definitions (Portuguese), which supports WCAG 3.1 intent for unusual words and abbreviations.

- *Semantic integrity gate (level-aware)*: Before returning, the output is validated to preserve essential legal information. In `light`, integrity is strict and aims to preserve nearly all numeric and legal references. In `medium/strong`, integrity is configured for summarization: it still preserves dates, values, percentages, legal citations and registry/process references, but allows omission of non-essential personal qualifications (e.g., CPF, RG, address) when they do not affect the legal meaning of the excerpt.
- *Extension integration*: The MV3 manifest injects a content script and exposes `sidebar.css` as a web-accessible resource. The popup selects `level` and communicates via tab messaging. For PDFs, the prototype supports (a) forwarding Base64 through `/simplify` and (b) forwarding raw bytes through `/simplify-pdf`. For `local file://` PDFs, the user may need to explicitly allow file access for the extension in the browser settings.

Selected normalization rules (examples). Table 3 lists representative normalization issues and where in the pipeline they are handled.

Table 3: Examples of normalization and where they are applied.

Issue	Example (in $\rightarrow$ out)	Origin
Dash break in numbers	133- \n 569 $\rightarrow$ 133-569	PDF extraction fix
PT-BR decimals spacing	133.569, 32 $\rightarrow$ 133.569, 32	Preprocess regex
Remove digit gaps	1 2 3 $\rightarrow$ 123	Preprocess regex
Currency spacing	R\$133 $\rightarrow$ R\$ 133	Preprocess regex
Abbreviation shielding	Art. 5° (no split)	Splitter guard

Source: prepared by the author (2026)

Server hardening and warm-up. The API enables CORS, increases keep-alive timeouts, and performs a best-effort warm-up (`/api/tags + 1-token chat`) to reduce the first-call penalty while not blocking availability if the daemon is cold.

4.4 Limitations

The current version of LASCAS should be understood as a local-first proof of concept, and several limitations follow directly from this design. Because extraction, preprocessing, validation, and simplification run on the user’s own machine, performance depends on available hardware and becomes more sensitive with long legal documents and large PDFs. This constraint also influenced the adoption of a lightweight local model configuration, which helps keep the system feasible on commodity machines but limits performance and domain specialization. The performance benchmark reported in Section 6 quantifies this constraint on a concrete consumer-grade configuration.

In addition, the underlying language model was not specifically fine-tuned on a broad corpus of Brazilian legal simplification pairs, which means that part of the system’s quality still relies on prompt design, preprocessing, integrity checks, and fallback rules. As a result, the output may occasionally miss stylistic or terminological nuances typical of highly formal legal and notarial documents.

There are also practical limitations in document handling and accessibility support. The glossary layer improves comprehension, but it is finite and manually curated, so rare expressions, regional usages, or institution-specific abbreviations may still remain difficult. PDF processing is also more reliable when files contain an actual text layer; scanned or image-only PDFs are not handled with the same robustness because the prototype does not yet include a dedicated OCR stage. Finally, recurring headers, footers, and institutional metadata remain a challenging case, since repeated elements may be either noise or legally relevant content. Even with targeted heuristics, future versions should improve parsing, OCR support, glossary coverage, and domain adaptation to reduce reliance on case-specific adjustments.

5 EVALUATION METHODOLOGY

We adopt a mixed attitudinal approach that combines perceived usability (via the System Usability Scale, SUS) with acceptance constructs (via the Technology Acceptance Model, TAM). This choice is supported by recent SUS benchmarking work in applied settings and systematic evidence of TAM’s suitability for technology-adoption research (HYZY et al., 2022; ROSLI et al., 2022). Our design also follows studies that explicitly integrate SUS and TAM when evaluating interactive systems (CHUENYINDEE et al., 2022).

5.1 Participants and Procedure

Participants use the LASCAS browser extension to simplify at least one real legal document (HTML or PDF) and then complete the questionnaire. The procedure is unmoderated and consists of three steps: brief task instructions, task execution with the extension, and a post-task questionnaire (Likert items plus one open-ended question).

5.2 Instrument: Likert Scale and Items

All items use a 5-point Likert scale (1 = *Strongly disagree*, 2 = *Disagree*, 3 = *Neither*, 4 = *Agree*, 5 = *Strongly agree*). The ten statements are **randomized** to avoid presenting the positive/negative pair of the same construct consecutively, an arrangement recommended to reduce response artifacts in lightweight usability evaluations (ROBERTSON; KORTUM, 2020). The complete list of items is shown in Table 4. The statements cover five constructs: Perceived Ease of Use (PEOU), Learnability, Perceived Usefulness (PU), Performance (latency/throughput),

and Behavioral Intention (BI).

Table 4: Survey items in randomized order (negative items are reverse-scored).

ID	Statement (English)	Dimension	Polarity
Q1	The extension is easy to use.	PEOU (SUS/TAM)	Positive
Q2	I learned to use the extension quickly.	Learnability (SUS)	Positive
Q3	The system’s responses are fast enough for my tasks.	Performance (pragmatic)	Positive
Q4	I intend to keep using this extension in my work or studies.	BI (TAM)	Positive
Q5	The extension helps me understand legal text more effectively.	PU (TAM)	Positive
Q6	I needed to learn a lot before I could get going with this extension.	Learnability (SUS)	Negative (reverse)
Q7	The system is often too slow for me to be productive.	Performance (pragmatic)	Negative (reverse)
Q8	The extension is not easy to use.	PEOU (SUS/TAM)	Negative (reverse)
Q9	The extension does not improve my understanding of legal text.	PU (TAM)	Negative (reverse)
Q10	I do not intend to use this extension in the future.	BI (TAM)	Negative (reverse)

Source: prepared by the author (2026)

At the end of the questionnaire, we include the following open-ended prompt:

In your own words, what worked well and what should be improved in the extension?

5.3 Scoring and Data Handling

This subsection outlines the scoring pipeline used to derive construct-level indices and an optional SUS-style composite. We first define the coding scheme, then compute construct means, perform an inconsistency check, assess reliability, and specify reporting outputs.

- *Coding:* Responses are mapped to integers 1–5 in ascending agreement. Negative items (Q6, Q7, Q8, Q9, Q10) are reverse-coded via $\text{rev}(x) = 6 - x$. This scoring is consistent with contemporary SUS/TAM practice in mixed instruments (ROBERTSON; KORTUM, 2020).
- *Construct scores:* We compute per-construct means using the positive/negative pairs in

their randomized positions:

$$\begin{aligned} \text{PEOU} &= \text{mean}(Q1, \text{rev}(Q8)), & \text{Learnability} &= \text{mean}(Q2, \text{rev}(Q6)), \\ \text{PU} &= \text{mean}(Q5, \text{rev}(Q9)), & \text{Performance} &= \text{mean}(Q3, \text{rev}(Q7)), \\ \text{BI} &= \text{mean}(Q4, \text{rev}(Q10)). \end{aligned}$$

For the SUS-style composite, after reverse-coding the negative items, the ten item scores (each in the 1–5 range) are summed, producing a raw score from 10 to 50. This sum is then linearly rescaled to a 0–100 scale using the following equation:

$$\text{Composite} = \left(\sum_{i=1}^{10} \text{item}_i - 10 \right) \times 2.5.$$

This follows the same rescaling logic as the original SUS, adapted to the item set used in this study (HYZY et al., 2022).

- *Inconsistency check:* For each pair, we compute $\Delta = |Q^+ - \text{rev}(Q^-)|$; large discrepancies (e.g., $\Delta > 2$) are flagged as potential inattentive responses. Flags are reported as a data-quality indicator and not used for automatic exclusion.
- *Reliability and analysis:* We report descriptive statistics (mean, standard deviation, median, and favorable response rates) for each construct and for the SUS-style composite. Internal consistency is reported for the full scale using Cronbach’s α , while pairwise construct reliabilities are interpreted cautiously because each construct is represented by only two items. The TAM lens supports interpretation of PEOU and PU as relevant dimensions for understanding Behavioral Intention in the analysis narrative.
- *Outputs and Reporting:* We present: (i) construct means for PEOU, Learnability, PU, Performance, and BI; (ii) the optional SUS-style composite (0–100); (iii) the proportion of responses flagged by the inconsistency check; and (iv) key themes from the open-ended answers. This structure aligns with recent integrated SUS+TAM evaluations in educational and productivity technologies.

5.4 Performance Benchmark

To complement the attitudinal instrument, in which performance appears only as a perceived construct (Q3/Q7), we also measured the objective runtime behavior of the backend. The benchmark reproduces the main condition of the broader study: the public petition most frequently assessed by participants (4,392 extracted words) is simplified at the `medium` level through the same `/simplify-pdf` endpoint used by the extension. After a priming request that keeps model loading out of the measurements, 10 identical warm repetitions are executed by a self-contained script kept in the project repository, with no changes to the pipeline: processing

telemetry comes from the instrumentation the API already returns, end-to-end latency is measured on the client side, and GPU memory, GPU utilization, and RAM are sampled at 1-second intervals. The reported metrics are end-to-end latency, PDF extraction and transport time, normalized input throughput (words per second; seconds per 1,000 words), and the resource-usage envelope. Table 5 summarizes the hardware and software configuration, a mid-range consumer laptop consistent with the local-first premise of the prototype.

Table 5: Hardware and software configuration used in the performance benchmark.

Component	Configuration
Machine	Alienware m16 R1 (laptop)
CPU	Intel Core i9-13900HX (32 logical processors)
RAM	31 GB available to the host OS; 15.5 GB provisioned to WSL2
GPU	NVIDIA GeForce RTX 4070 Laptop GPU, 8 GB VRAM (driver 610.88)
OS / environment	Windows 11 Pro with WSL2 (Ubuntu 26.04 LTS)
Backend runtime	Node.js v24.19.0; Ollama 0.32.7
Language model	Llama 3.1 8B instruct (Q4_K_M, 8,192-token context, system prompt embedded via Ollama Modelfile)

Source: prepared by the author (2026)

6 RESULTS AND DISCUSSION

This section presents the empirical results obtained with LASCAS and discusses them in light of the goals established in Sections 4 and 5. Rather than treating evaluation as a purely benchmark-driven exercise, this work adopted an applied and iterative validation strategy centered on real legal documents, human perception, and cognitive accessibility. This framing is coherent with the practical character of the prototype and with the data-scarcity problem already identified in related work on legal simplification (GARIMELLA et al., 2022; ALVES et al., 2023; ATHUGODAGE; MITROFANOVA; GUDKOV, 2024).

6.1 Evaluation Scope and Rationale

The empirical stage was organized in two complementary moments with human participants, complemented by the automated backend performance benchmark defined in Section 5. First, a formative pre-test was conducted with three participants from the legal domain using three public legal files that were anonymized for the study. This smaller round had an exploratory role: it helped verify whether the prototype had reached a satisfactory maturity threshold for a broader trial, identify which failure modes still required attention, and define a feasible evaluation format. A key outcome of this pre-test was the decision to adopt the `medium`

simplification level in the larger study, since it offered the best balance between readability gain, semantic preservation, and task duration.

Second, a broader attitudinal evaluation was conducted with the questionnaire described in Section 5. In this stage, each participant assessed one legal document simplified at the `medium` level. This decision intentionally reduced reading burden and kept the session short enough to avoid fatigue, while still allowing participants to compare the original and simplified versions with adequate attention. The formative pre-test with three legal-domain participants was therefore used as a calibration step, not as a statistically conclusive comparison between all levels. Running `light`, `medium`, and `strong` in the broader study would have substantially increased reading time and introduced fatigue effects, since each participant would need to inspect multiple versions of the same legal content. For this reason, the broader evaluation focused on the level that had shown the best practical trade-off in the pre-test, while a controlled comparison between simplification levels remains a direction for future evaluation. Table 6 summarizes the evaluation scope.

Table 6: Evaluation stages and rationale adopted in this study.

Stage	Primary goal	Material	Participants	Main outcome
Formative pre-test	Calibrate the prototype before broader exposure	3 public legal documents, anonymized for the study; additional real files used internally during development.	3 (legal domain)	Selection of the <code>medium</code> level as the best overall trade-off; refinement of prompts, glossary behavior, and document-handling heuristics.
Broader questionnaire study	Measure perceived usability, usefulness, and adoption potential	1 anonymized document per participant, presented in original and medium-simplified form.	30	Quantitative SUS/TAM-style evidence plus open-ended feedback on strengths and improvement opportunities.
Backend performance benchmark	Characterize objective runtime behavior on consumer hardware	The petition most used in the broader study, at the <code>medium</code> level, 10 warm repetitions.	n/a (automated)	Latency, throughput, and resource envelope; model inference confirmed as the dominant computational cost.

Source: prepared by the author (2026)

It is important to stress that the number of publicly shareable files was intentionally limited. During development, more than ten real legal files were used to stress the extraction, preprocessing, validation, and simplification pipeline; however, only three public documents could be kept

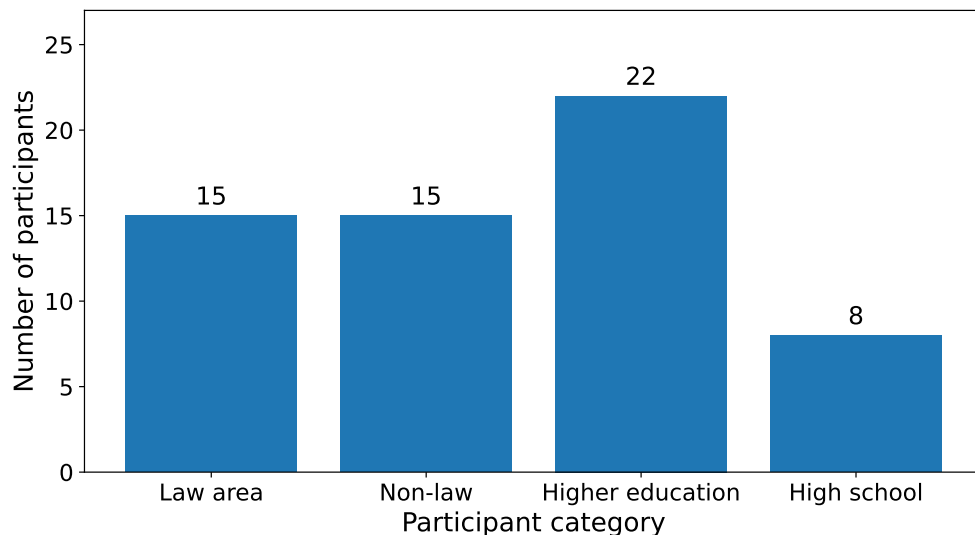
fully available in a reproducible way for download and inspection. In the legal domain, there is still no broadly accepted benchmark composed of full original legal documents paired with validated simplified counterparts. For this reason, the evaluation strategy privileged ecological validity: authentic documents, anonymization, and iterative feedback from people familiar with legal practice.

This also helps explain why development was not pushed indefinitely beyond the current point. Going much further with targeted refinements would start steering the solution toward the specific test files, undermining the generic orientation of the system. Since a central design decision of LASCAS is to remain *generalist*, the prototype was considered sufficiently mature for this stage once it reached stable behavior across heterogeneous documents and once the pre-test indicated that the medium level was already producing a satisfactory balance between condensation and preservation.

6.2 Participant Profile

The broader questionnaire stage included 30 respondents. The sample was intentionally heterogeneous in relation to legal background, with perfect balance between participants from the legal area ($n = 15$) and participants from outside the legal area ($n = 15$). In terms of formal education, 22 participants reported completed higher education and 8 reported completed high school. Figure 5 summarizes this profile.

Figure 5: Participant profile in the broader questionnaire stage.



Source: prepared by the author (2026)

This composition is relevant because it brings together two audiences that matter for the problem addressed here. Participants with legal-domain familiarity help indicate whether the

simplified output remains dependable as an initial reading aid, while non-specialists help show whether the system actually lowers the entry barrier created by legal language. For a tool whose purpose is to improve accessibility, positive feedback distributed across both groups is more meaningful than approval restricted to expert users alone. The balanced split also supports the subgroup reading presented later in this section, since it allows the same descriptive indicators to be compared across profiles of equal size. Educational background plays a similar role: because part of the sample did not complete higher education, the evaluation includes readers closer to the literacy profiles that motivate cognitive accessibility work in the first place. This does not turn the sample into a clinical population, but it brings the test conditions closer to the audiences the prototype intends to serve.

6.3 Quantitative Results

The quantitative results show a consistently favorable pattern across different dimensions of use. As defined in Section 5, negative items were reverse-scored before construct calculation, and the data were examined both at construct level and at item level. Taken together, the results indicate that the prototype was perceived as easy to use, useful for initial reading of legal texts, and plausible as support for comprehension and triage tasks.

6.3.1 Construct-level outcomes

Table 7 presents the descriptive statistics by construct. On the 1–5 scale, the highest means were observed for Perceived Ease of Use and Learnability (both 4.68), followed by Performance (4.65) and Perceived Usefulness (4.63). Behavioral Intention was the lowest-scoring construct (4.43), but it still remained clearly favorable, with 86.7% of respondents at or above 4.0.

Table 7: Construct-level scores in the broader questionnaire stage.

Construct	Mean	SD	Median	Favorable (% ≥ 4)
PEOU	4.68	0.40	5.0	96.7
Learnability	4.68	0.38	5.0	100.0
Performance	4.65	0.53	5.0	96.7
PU	4.63	0.54	5.0	93.3
BI	4.43	0.65	4.5	86.7

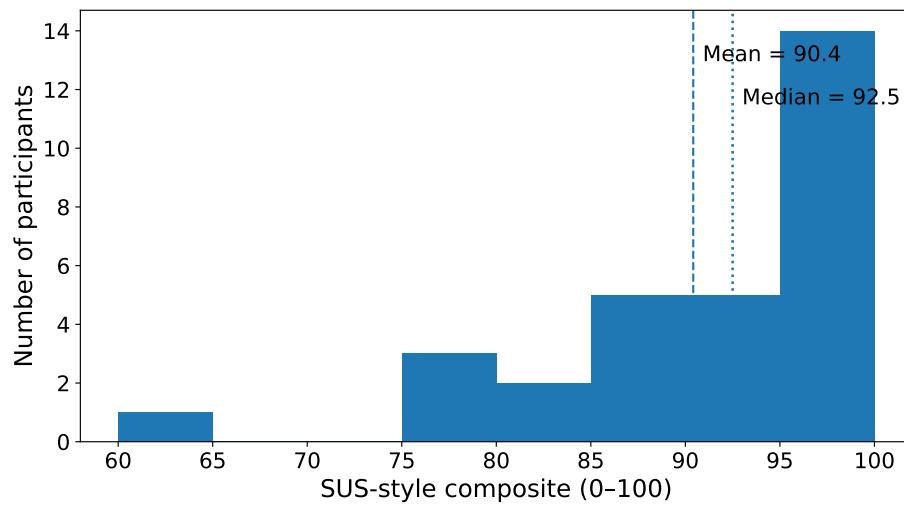
Source: prepared by the author (2026)

These construct means concentrate in the upper range of the scale, suggesting that the prototype was not merely considered promising in the abstract. It was already perceived as understandable, learnable, and useful in a concrete task involving legal documents. For an

accessibility-oriented system, this matters because a simplified output has limited value if the interaction required to obtain it becomes a new source of friction.

The overall SUS-style composite reinforces this interpretation. After linearly rescaling the ten reverse-aligned items, the instrument reached a mean of 90.4/100 ($SD = 9.8$), with median 92.5 and interquartile range from 87.5 to 97.5. In practical terms, this is a strong result for a local-first academic prototype. Moreover, 86.7% of participants scored the experience at 80 or above, and 63.3% scored it at 90 or above. Figure 6 shows the distribution.

Figure 6: Distribution of the SUS-style composite (0–100).



Source: prepared by the author (2026)

The quality checks described in Section 5 also support a cautious but positive reading of the dataset. Only 2 respondents (6.7%) triggered at least one inconsistency flag above the adopted threshold. Since these flags were not used for exclusion, the analysis remains conservative. At the same time, the low flagged proportion indicates that the questionnaire was completed with reasonable coherence, which increases confidence in the descriptive interpretation of the results. Full-scale internal consistency was also good ($\alpha = 0.832$). Pairwise construct reliabilities should be interpreted carefully because each construct was measured with only two items and the response distribution shows a visible ceiling effect; even so, the pattern is informative and especially favorable for the Performance pair ($\alpha = 0.784$).

The ceiling effect should also be considered when interpreting the strength of the results. Because most construct means were concentrated between 4.43 and 4.68 on a 1–5 scale, the instrument had limited room to discriminate between moderately positive and extremely positive perceptions. Therefore, the findings should be read primarily as evidence of strong perceived acceptance and usability in this evaluation context, not as a fine-grained ranking of constructs or subgroups. Future evaluations should include more discriminative measures, such as objective comprehension tasks, task-completion time, error analysis, and larger samples, to better

separate usability, understanding, and adoption effects.

6.3.2 Item-level interpretation

Table 8 presents the aligned item-level statistics. All ten items obtained means above 4.3 after reverse coding, and the lowest favorable rate was 90.0%, observed for the two Behavioral Intention items. This helps show that the positive evaluation was not concentrated in a single isolated aspect of the experience.

Table 8: Item-level scores after reverse coding of negative items.

Item	Mean	SD	Favorable ($\% \geq 4$)
Q1	4.77	0.43	100.0
Q2	4.77	0.43	100.0
Q3	4.83	0.46	96.7
Q4	4.50	0.68	90.0
Q5	4.77	0.50	96.7
Q6	4.60	0.50	100.0
Q7	4.47	0.68	96.7
Q8	4.60	0.56	96.7
Q9	4.50	0.86	93.3
Q10	4.37	0.89	90.0

Source: prepared by the author (2026)

One detail is worth noting here. The slightly more conservative scores were associated with future-oriented adoption rather than with immediate usability or perceived usefulness. This is a realistic pattern for an early-stage legal technology tool: users can recognize practical value before they are fully ready to imagine routine dependence on it. In that sense, the lower Behavioral Intention scores do not weaken the proposal; they help place it at a plausible stage of maturity.

6.3.3 Subgroup stability

The favorable pattern was not restricted to a single user profile. Table 9 summarizes the SUS-style composite and construct means by subgroup. Legal-area participants reported slightly higher scores overall, but the differences were small enough to suggest stability rather than segmentation. The same is true for education level: even participants with completed high school evaluated the system positively across all measured dimensions.

Table 9: Subgroup results for the SUS-style composite and construct means.

Group	n	SUS-style	PEOU	Learnability	PU	BI
Legal area	15	91.3	4.77	4.73	4.70	4.53
Non-legal area	15	89.5	4.60	4.63	4.57	4.33
Higher education	22	90.6	4.70	4.70	4.68	4.43
High school	8	89.7	4.63	4.63	4.50	4.44

Source: prepared by the author (2026)

This subgroup stability is important for the broader argument of the thesis. A legal simplification system that performs well only for readers who are already comfortable with legal discourse would have limited accessibility value. Here, the descriptive evidence points in the opposite direction: the perceived benefit appears to extend beyond the specialist audience.

6.4 Qualitative Feedback

The open-ended responses add important texture to the numerical picture. A light thematic reading of the 30 comments identified four recurring themes: clearer understanding and easier reading, productivity gains and faster triage, practicality of the interface, and requests for more detail, formatting, or glossary support.

The coding was descriptive rather than exhaustive, focusing on the perceptions that most directly help interpret the quantitative pattern. This is appropriate for the scope of the study, since the questionnaire was designed mainly to validate perceived usefulness and usability while still leaving space for participants to point out concrete strengths and improvement opportunities. Table 10 summarizes these frequencies as coded mentions. Because a single response could be coded into more than one theme, totals should be interpreted as theme occurrences rather than unique participants.

Table 10: Frequency of coded mentions in open-ended feedback.

Theme	Coded mentions
Clearer understanding and easier reading	17
Productivity, triage, and quick overview	15
Practicality, interface, and ease of use	12
Requests for more detail, formatting, or glossary support	9

Source: prepared by the author (2026)

The qualitative feedback aligns closely with the quantitative results. Many participants de-

scribed the extension in terms of orientation, overview, and first understanding, which suggests that LASCAS is not being interpreted as a replacement for full legal reading. Instead, it is seen as a support layer that lowers the initial barrier to entry into the document. This is precisely the role expected from a cognitively oriented accessibility tool. The co-occurrence of comprehension and productivity themes reinforces this reading: for many respondents, understanding the document better and dealing with it faster were not competing benefits, but two effects of the same reduction in cognitive load. Comments about practicality and ease of use, in turn, echo the high PEOU scores in spontaneous form, strengthening the construct-level interpretation with unprompted evidence.

One of the most illustrative comments reported a perceived reduction in reading time from approximately eight minutes on the original file to four minutes with the support of the extension. Although this is anecdotal rather than experimental timing evidence, it captures a practical advantage that appears repeatedly across the responses: the tool seems useful not only for understanding more, but for understanding earlier. Other comments reinforce this idea by connecting the prototype to triage, quick consultation, high-volume reading routines, and communication with clients in more accessible language.

An equally relevant aspect is the nature of the suggested improvements. Participants mostly asked for refinement rather than redesign: more detail in specific passages, better output formatting, richer glossary support, and, in some cases, even faster processing. This type of feedback is valuable because it suggests that the core proposition was accepted and that the next stage is one of quality improvement rather than basic justification. It is also notable that the suggestions concentrate on presentation, coverage, and speed rather than on the dependability of the simplified content, an encouraging signal for a tool operating in a domain where trust is decisive.

6.5 Backend Performance

Table 11 consolidates the benchmark defined in Section 5. The ten warm repetitions were remarkably stable: end-to-end latency averaged 153.5 s (SD 1.1 s; median 153.1 s; range 153.0–156.4 s), a coefficient of variation below 1%, which corresponds to 28.6 input words per second, or approximately 35 s per 1,000 words, at the `medium` level. PDF extraction and transport accounted for 54 ms on average, below 0.1% of the total, so virtually the entire cost of a request lies in the simplification core and, specifically, in the local model calls: each repetition issued 10 calls over 5 chunks, with no timeouts, at an average of roughly 15 s per call. A short priming request of 45 words completed in 4.3 s, indicating that per-request overhead is small once the model is resident and that latency is dominated by generation over long inputs rather than by fixed costs.

Table 11: Backend performance at the `medium` level (10 warm runs, 4,392-word petition).

Metric	Result
End-to-end latency (mean $\pm$ SD)	153.5 $\pm$ 1.1 s
End-to-end latency (median; min–max)	153.1 s; 153.0–156.4 s
PDF extraction and transport (mean)	54 ms (<0.1% of total)
Input throughput	28.6 words/s ($\approx$ 35 s per 1,000 words)
Model calls per run	10 across 5 chunks (no timeouts)
Mean GPU utilization	94.1% (per-run means 92.6–95.3%; peaks at 100%)
Peak GPU memory (global)	6,657 MiB of 8,188 MiB
Sustained GPU power	88–91 W
Peak RAM (WSL2)	$\approx$ 5 GB

Source: prepared by the author (2026)

The resource telemetry gives concrete form to the hardware sensitivity acknowledged in Section 4. Mean GPU utilization stayed around 94% (per-run means between 92.6% and 95.3%, with peaks at 100%) throughout every run, global GPU memory peaked at 6,657 MiB of the 8,188 MiB available with the quantized model resident, and board power was sustained around 88–91 W, while the CPU-side footprint stayed modest (peak of about 5 GB of RAM in the WSL2 environment). In practical terms, the accelerator is nearly saturated during the entire simplification window: the configuration of Table 5 is sufficient, but with limited headroom, and weaker GPUs, smaller VRAM budgets, or CPU-only execution can be expected to push latency well beyond the values reported here. This measurement turns the hardware requirement, previously stated qualitatively, into an explicit operating envelope for the local-first premise.

Finally, the objective numbers help contextualize the perceived-performance results. Participants rated the pace of the system very favorably (Q3 was the highest-scoring item, at 4.83, and the Performance construct averaged 4.65) even though a full-document pass takes about two and a half minutes on this configuration. This is coherent with the triage-oriented usage pattern identified in the qualitative feedback: a single wait of a few minutes replaces a longer unassisted reading effort, as in the report of reading time dropping from roughly eight to four minutes. The benchmark is intentionally scoped to the main study condition, one document at the `medium` level on one machine; characterizing scaling across document sizes, simplification levels, and hardware tiers remains future work.

6.6 Discussion and Positioning in the Literature

The results position LASCAS in an interesting place relative to the studies reviewed in Section 3. Prior work has consistently highlighted three central tensions in legal simplification:

the scarcity of parallel data, the difficulty of evaluating simplification quality, and the risk of semantic loss. LASCAS does not resolve these tensions once and for all, but the empirical evidence suggests that its architecture offers a productive way of navigating them in practice.

A first contribution is operational. While several related studies focus primarily on training strategies or corpus construction, LASCAS contributes a full prototype perspective. The system is not only a simplification model; it is a document-oriented workflow that includes extraction, legal-aware normalization, level-aware simplification, integrity validation, glossary support, and browser-side presentation. For accessibility purposes, this matters because users experience the whole interaction chain, not only the local quality of sentence rewriting.

A second contribution concerns evaluation. As already suggested by Garimella et al. (2022), automatic metrics and human judgment do not necessarily move in parallel in legal simplification. The present study extends that point by showing that user-centered evidence can be organized around perceived clarity, usefulness, and workflow fit, rather than around simplification scores alone. In this sense, the contribution of LASCAS is not to replace metric-oriented research, but to complement it with evidence of value in realistic use. The performance benchmark adds an objective layer to this argument: the studies reviewed in Section 3 concentrate on output quality and do not characterize runtime feasibility on end-user hardware, while the local-first premise of LASCAS makes this dimension unavoidable. Reporting latency, throughput, and resource usage alongside perceived usability therefore broadens the evidence base on which adoption decisions for legal simplification tools can be discussed.

A third contribution lies in how the prototype is positioned. Participants repeatedly framed the system as support for initial reading, triage, and communication, not as a substitute for specialized legal interpretation. This is a productive framing for the current stage of legal AI: tools of this kind add most value when they mediate access, not when they claim autonomous legal authority. This framing is also consistent with the human-oversight perspective discussed in Section 3, in which AI assistance is expected to augment rather than replace legal judgment (TOIVONEN; KETOLA; HAAPIO, 2025; ZIMINA-POIROT; GLEDHILL; BOUYÉ, 2025); LASCAS operationalizes that expectation by keeping the tool bounded to comprehension support, with the original document always in view. This positioning also helps explain why the prototype was evaluated as a complete workflow rather than as a benchmark-only simplification model. The relevant question is not only whether individual sentences become shorter, but whether the interaction helps users approach a legal document with clearer orientation and fewer barriers. In this respect, LASCAS complements model-centered work by emphasizing the practical conditions under which simplification becomes usable in a real reading situation. The subgroup stability reported earlier reinforces this mediating role, indicating that it is recognized by specialists and non-specialists alike. Table 12 positions LASCAS relative to the main studies discussed in Section 3.

Table 12: Positioning of LASCAS relative to related work.

Work	Main focus	Evaluation strategy	Relation to LASCAS
Garimella et al. (2022)	Simplification of English legal contracts	Human judgments and comparison with automatic metrics	Highlights the difficulty of aligning metrics with perceived simplicity; LASCAS advances the discussion through real-document use, integrity safeguards, and user-centered evidence.
Alves et al. (2023)	Simplification of Brazilian Portuguese judicial summaries	Readability metrics and model comparison	Demonstrates readability gains in the Brazilian context; LASCAS extends the discussion by integrating document processing, interface, and practical evaluation with users.
Athugodage et al. (2024)	Transfer learning with a newly built legal simplification corpus	Readability metrics and human evaluation	Shares the concern with data scarcity, but LASCAS emphasizes local execution and level-aware simplification instead of a new supervised corpus.
Toivonen et al. (2025) and Zimina-Poirot et al. (2025)	Plain language, legal design, and responsible AI assistance	Conceptual and case-based framing of accessibility in law	LASCAS operationalizes this agenda in a concrete artifact, showing how plain-language support can be embedded in a local-first simplification workflow.
LASCAS (this work)	Local-first legal simplification for cognitive accessibility	Formative pre-test plus broader SUS/TAM-style user evaluation on real anonymized files	Combines accessibility goals, document engineering, semantic-integrity safeguards, glossary support, and applied evaluation in Brazilian legal Portuguese.

Source: prepared by the author (2026)

Taken together, these findings suggest that the value of LASCAS lies less in competing with purely model-centered approaches and more in showing how simplification, accessibility, and real use can be articulated within the same artifact, complementing studies more narrowly focused on corpora, metrics, or training architecture.

7 CONCLUSION

This thesis investigated how contemporary Natural Language Processing techniques can be employed to simplify Brazilian Portuguese legal texts with cognitive accessibility as a guiding objective. In response to this problem, the work proposed and developed LASCAS, a **local-first** prototype capable of receiving legal documents in HTML or PDF, processing them, and presenting simplified versions through a browser-integrated interface.

The thesis contributes at three levels. Conceptually, it connects cognitive accessibility, plain

language, and automatic text simplification in a domain where insufficient comprehension can directly affect autonomy, informed decision-making, and access to legal information. Technically, it shows that legal simplification in Brazilian Portuguese benefits from a hybrid architecture in which the language model operates alongside preprocessing, protection of essential information, validation, and presentation layers. Empirically, it shows that this combination already yields a prototype perceived by users as useful, easy to operate, and relevant for real tasks involving initial reading and document triage.

The results suggest that the value of LASCAS lies not only in rewriting difficult passages, but also in reorganizing access to the legal document. Rather than competing with specialized legal reading, the tool acts as an intermediate support layer that helps users grasp the document’s central theme, function, and most relevant points more quickly.

This applied positioning is reinforced by the methodological stance of the work: the use of real documents, distinct participant profiles, and local execution conditions brought the evaluation closer to realistic contexts of use. The local-first design also made visible the practical trade-off between privacy, feasibility, latency, and model capacity, which is central for legal documents that may contain sensitive information. The backend benchmark made this trade-off concrete: on a consumer laptop GPU, a full petition was simplified in about two and a half minutes with the accelerator nearly saturated, showing that model inference defines the performance boundary of the approach. In this sense, the contribution of the prototype is practical as well as academic, since it demonstrates a feasible path for accessibility-oriented legal simplification beyond purely model-centered experiments.

From the perspective of the research field, the thesis reinforces that simplifying legal texts is not only a text-generation challenge, but also a mediation problem between specialized language and public understanding. It also suggests that accessibility in the legal domain depends not only on model quality, but on the broader interaction between document processing, semantic safeguards, and user-facing presentation. This broader view is especially relevant in legal contexts, where readability gains must be balanced with the preservation of essential meaning and trustworthy user support.

The study therefore points to a design direction in which legal simplification tools should be assessed not only by whether they produce simpler sentences, but also by whether they help users approach documents with more orientation, confidence, and control. Looking ahead, LASCAS opens directions for broader evaluations with objective comprehension tasks, richer glossary support, stronger PDF handling, OCR integration, latency optimizations for long documents, and models better adapted to Brazilian legal Portuguese.

In summary, LASCAS demonstrates that AI-assisted legal simplification can generate concrete gains in perceived clarity, usefulness, and accessibility without depending on excessively complex infrastructure. More than an isolated experiment, this work offers a solid basis for future evolution and supports the idea that language technologies can practically reduce the distance between legal documents and the people who need to understand them.

Its main contribution is therefore not a claim of full automation, but a practical demonstration that legal AI can be designed as a careful layer of support for comprehension. By keeping the system bounded to initial reading and triage, the work preserves the distinction between accessibility support and legal interpretation. This distinction is important for future development because it allows improvements in model quality, document coverage, and evaluation methods without changing the central role of the tool: helping users approach legal information with less friction and more autonomy.

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